

Durable Solutions

Information Sharing Session with Development Partners
HRRP, Kathmandu, June 2018

Agenda

- Durable Solutions – Phase II: Introduction
 - Geo-Hazard Assessments, Categorizations, Drawings & Updates
 - Risk Rating & Prioritization
 - DSCWM & DWIDM Interventions
 - NRA Geologists Activities
 - Consortium's Work Plans & Priorities
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Durable Solutions Consortium

Phase I (December 2016 – December 2017)



Phase II (February 2018 – February 2020)



Durable Solutions: Phase II

Project duration: February 2018 – February 2020

Facilitating the implementation of durable solutions for households at risk of or displaced by geo-hazards

Implementation & Coordination

Facilitating safe relocation of eligible households

Geo-hazard Assessments

Supporting NRA in surveying actual or potential geo-hazard sites

Policy Development Support

Technical assistance to NRA in developing & formulating new policies & guidelines

Building Resilience

Protection, shelter & livelihoods support

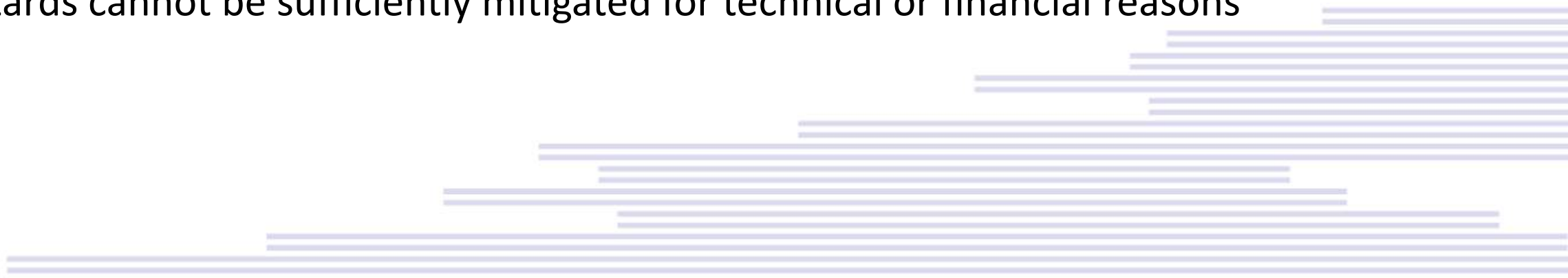


Geo-hazard Assessments & Categorization

Category 1 Households: Households whose livelihoods and/or shelters are at a safe location to begin reconstruction.

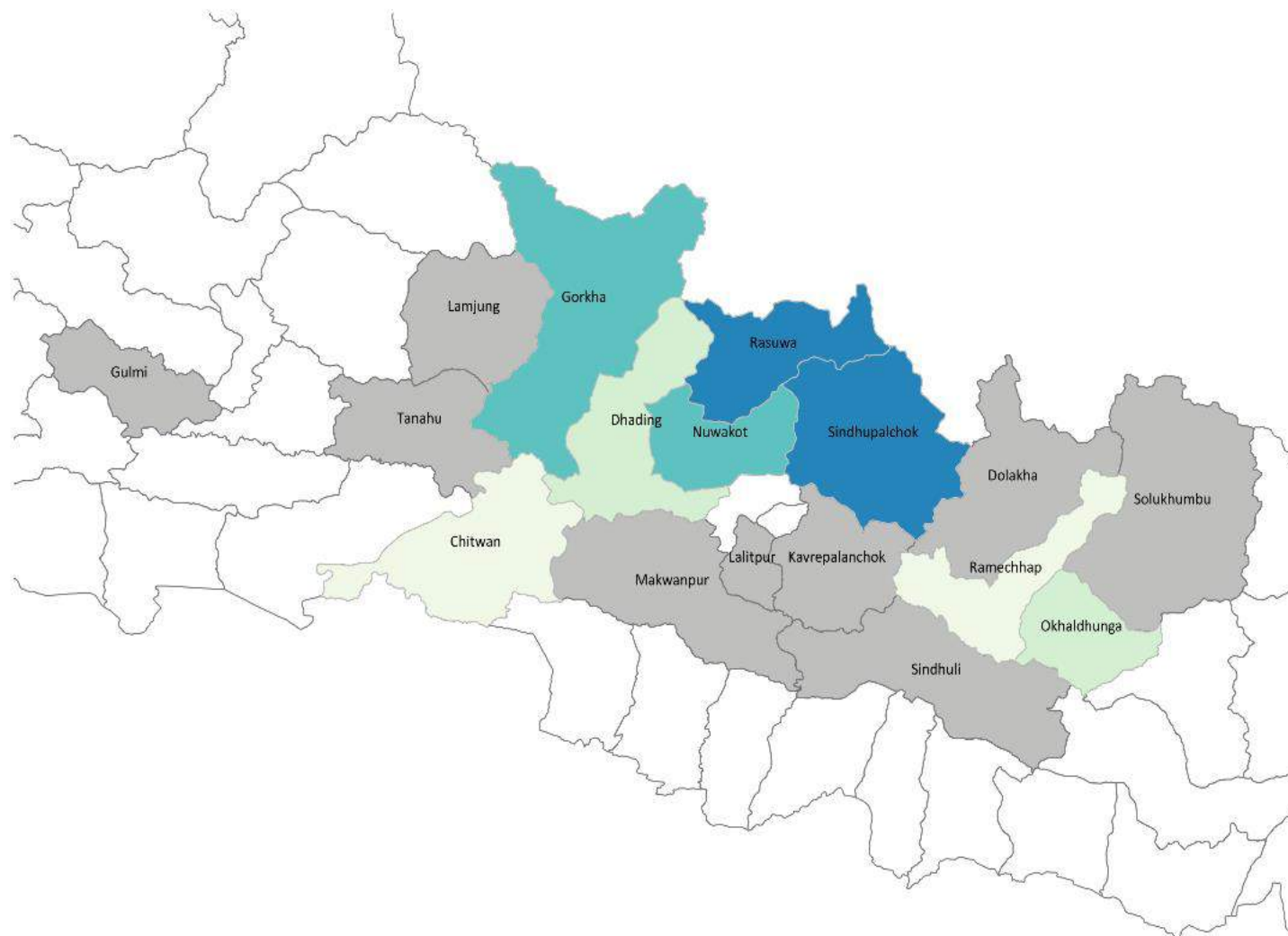
Category 2 Households: Households whose livelihoods and/or shelters are unsafe due to the existing state of geo-hazards, but the safety risk of these geo-hazards can be effectively mitigated.

Category 3 Households: Households whose livelihoods and/or shelters are unsafe due to the existing state of geo-hazards, and the safety risk of these geo-hazards cannot be sufficiently mitigated for technical or financial reasons

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Category 3 Households – Current Caseloads

Rasuwa	836
Sindupalchowk	688
Gorkha	368
Nuwakot	365
Dhading	259
Okhaldhunga	148
Ramechhap	150
Chitwan	102
Other	354
Total	3168

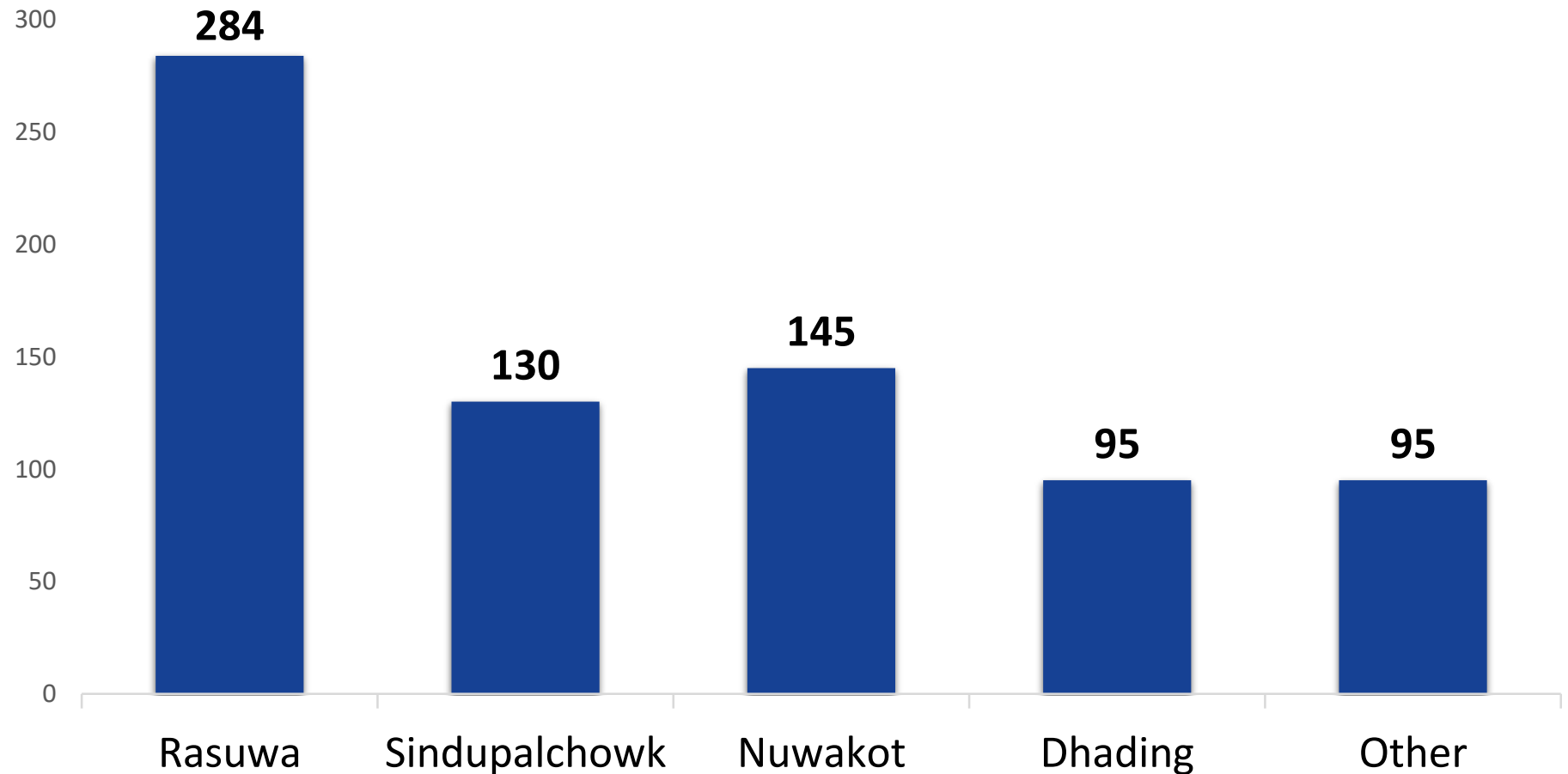


CAT-3 Households Relocated to date

749

**Relocated
Households
May 2018***

*Data available to date

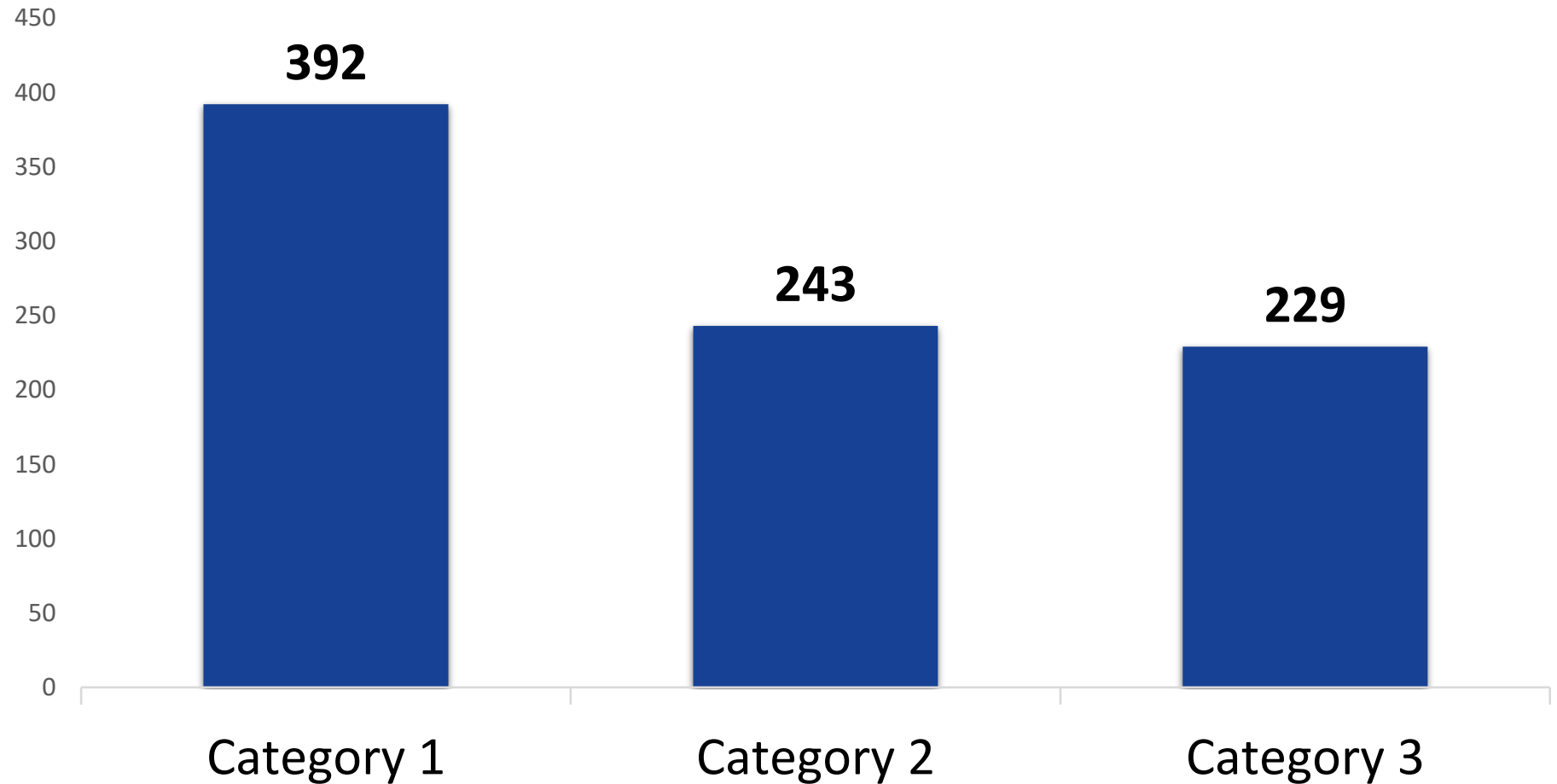


Geo-Hazard Assessments to date

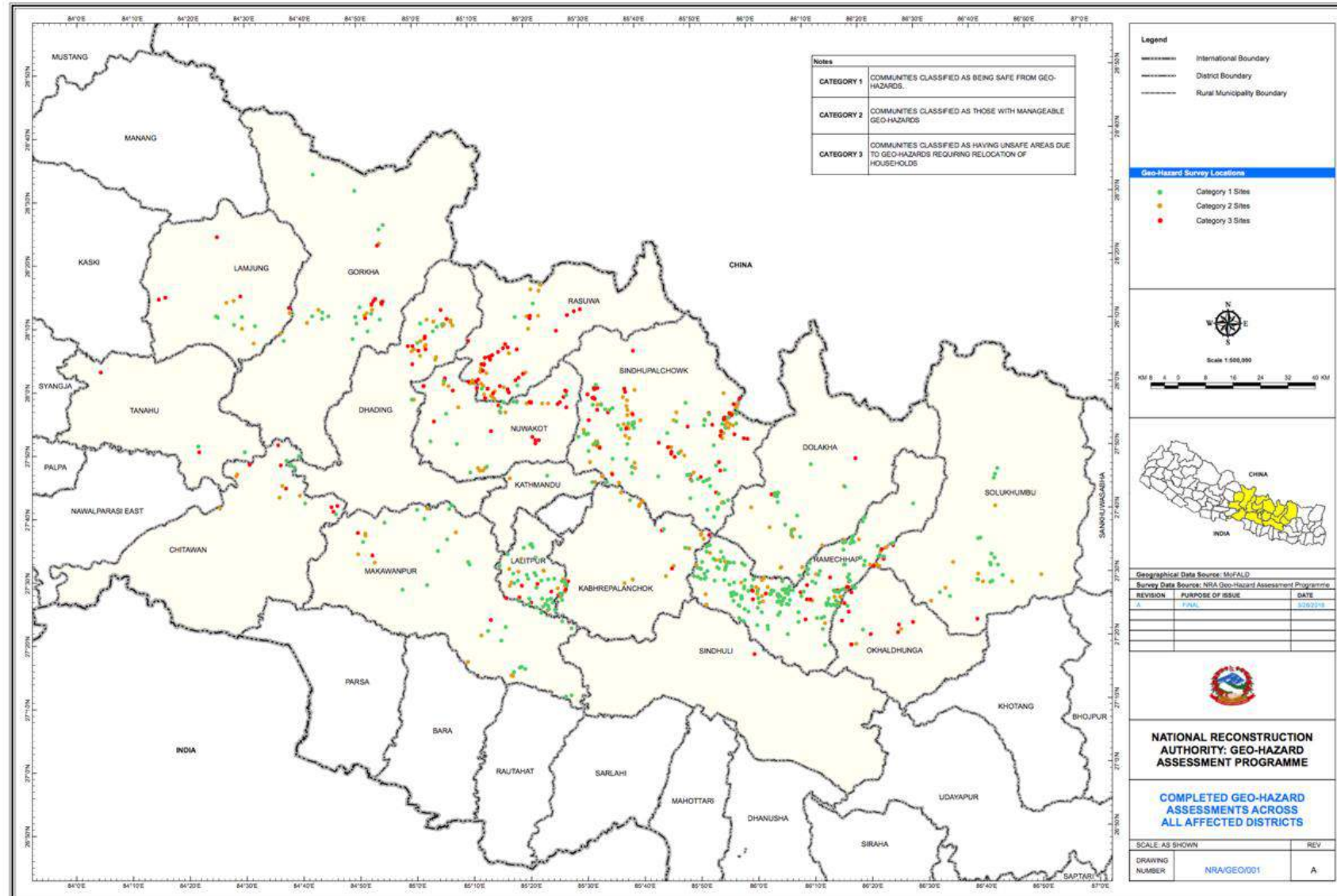
864

**Surveys
Conducted
May 2018***

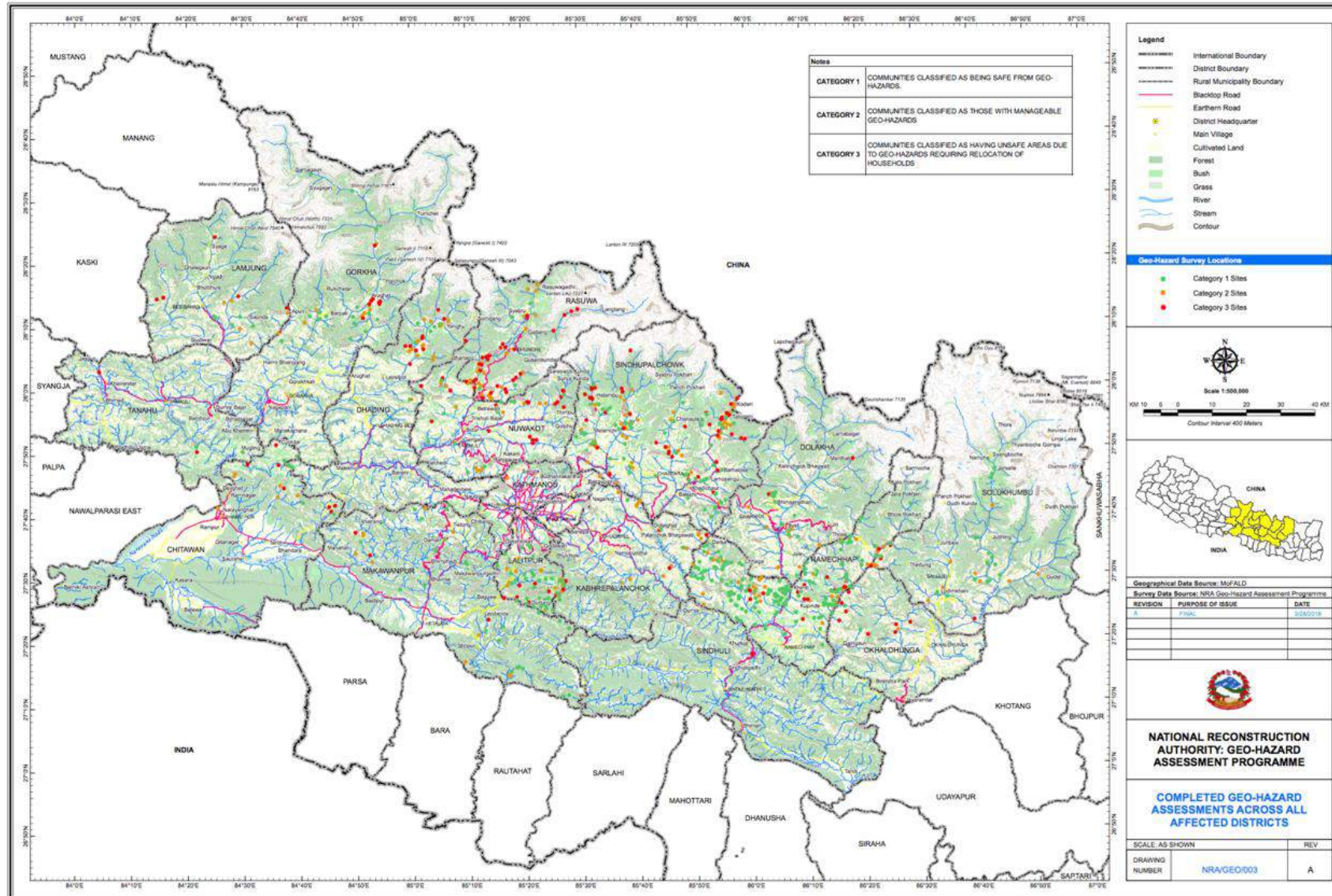
*Data available to date



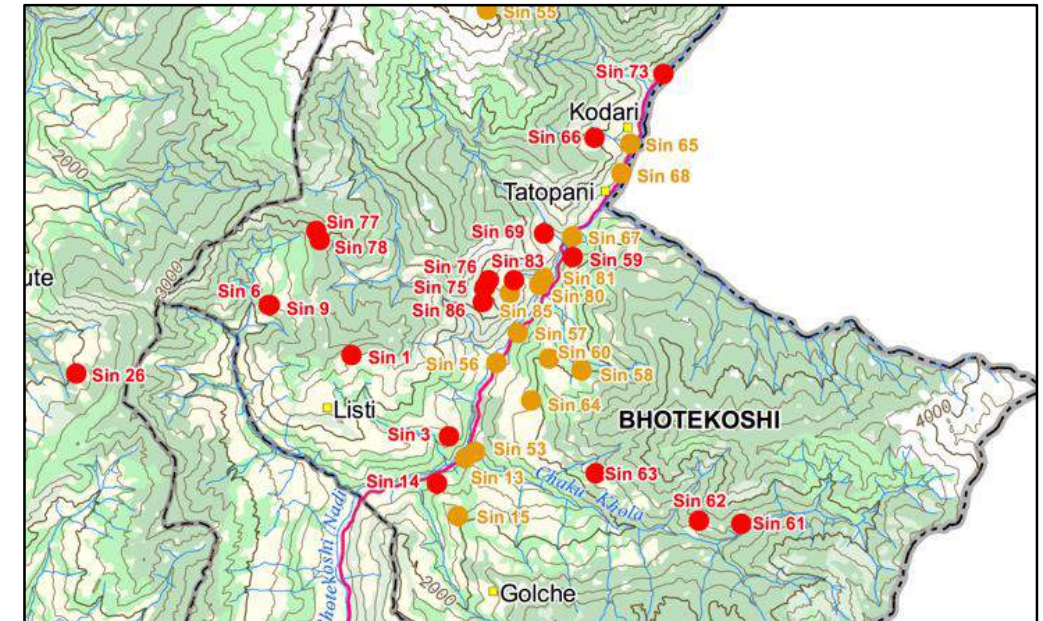
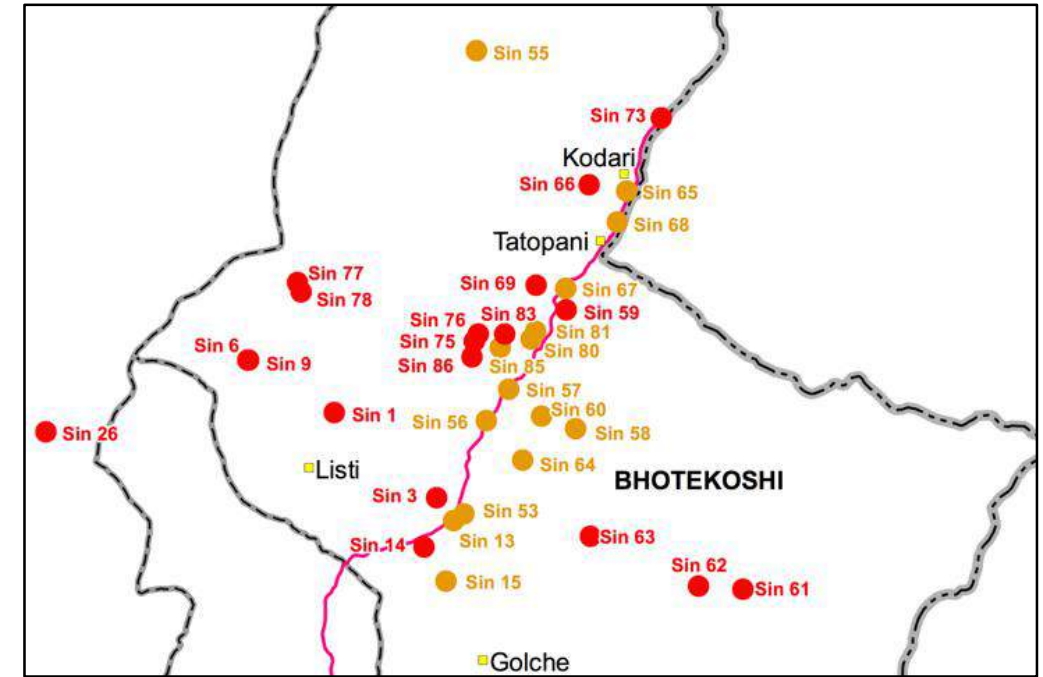
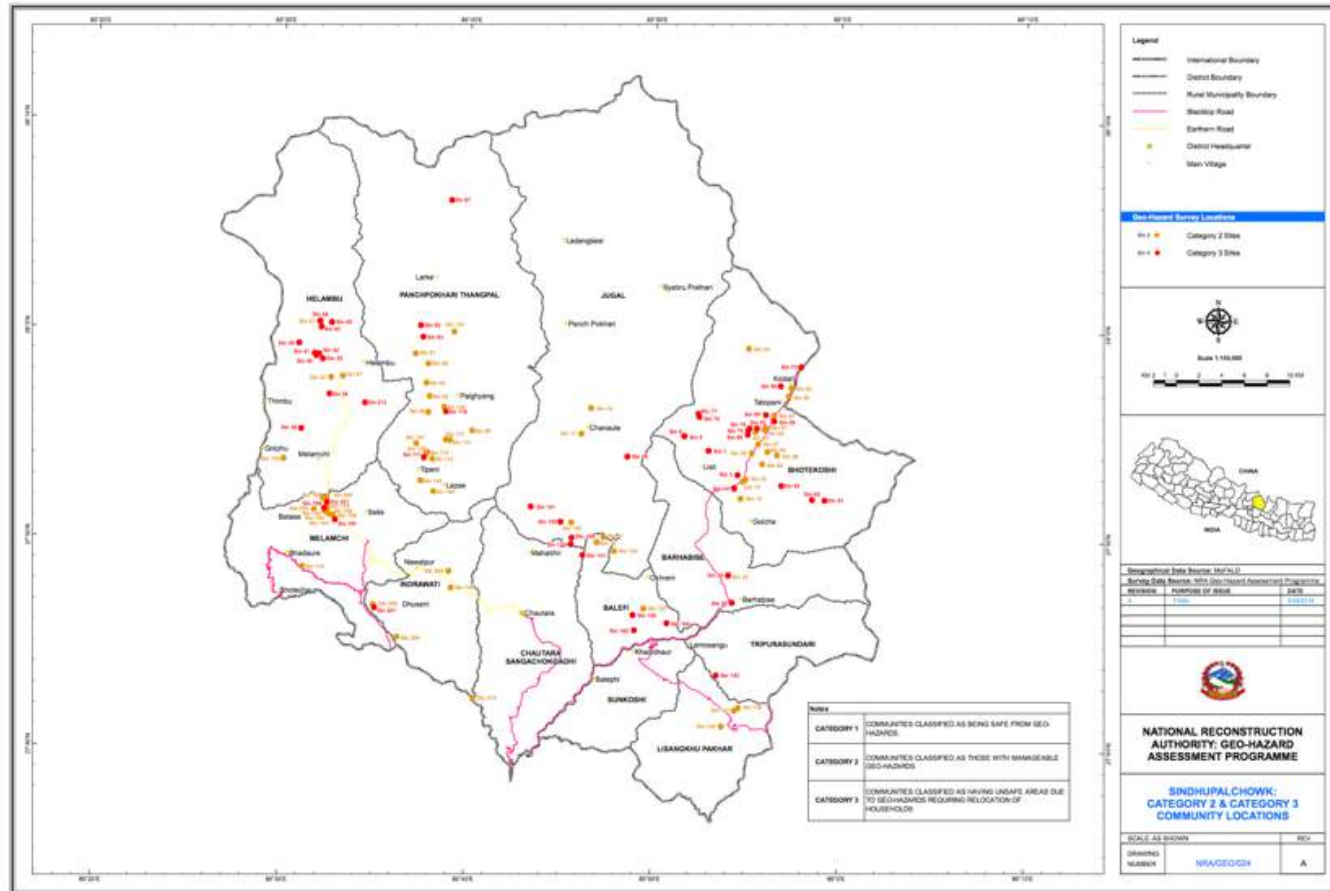
Drawings & References



Drawings & References



Drawings & References



Drawings & References

Ref	VDC	Ward	Settlement	Category 1	Category 2	Category 3	Easting	Northing	Remarks
Sin 60	Fulpingkatti	5	Salangkatti		1		395247	3087840	Gabion wall and bioengineering
Sin 61	Fulpingkatti	6	Yarmala-Hikudada			1	400269	3083515	18 HOUSEHOLDS TO BE RELOCATED
Sin 62	Fulpingkatti	6	Mukchyasa			1	399168	3083596	16 HOUSEHOLDS TO BE RELOCATED
Sin 63	Fulpingkatti	6	Chyasa			1	396467	3084847	4 HOUSEHOLDS TO BE RELOCATED
Sin 64	Fulpingkatti	7	Langkhola		1		394775	3086742	Due to road cutting - rock netting and bioengineering recommended.
Sin 65	Tatopani	1	Ghattekhola		1		397380	3093455	Gabion wall at road level, bioengineering.
Sin 66	Tatopani	2	Nadung			1	396440	3093616	3 HOUSEHOLDS TO BE RELOCATED
Sin 67	Tatopani	4	9Kilo		1		395855	3091030	Gabion wall at road level, bioengineering, rock netting
Sin 68	Tatopani	4	Koplang		1		397132	3092685	Gabion wall and bioengineering
Sin 69	Tatopani	5	Fugmache			1	395110	3091098	1 HOUSEHOLD TO BE RELOCATED
Sin 70	Tatopani	9	Bokchen/Bukchang	1			397054	3090831	
Sin 71	Tatopani	9	Tartung	1			397322	3091115	
Sin 72	Tatopani	9	Maikharka	1			398897	3091421	
Sin 73	Tatopani	3	Liping			1	398236	3095281	ALL 70 HOUSEHOLDS TO BE RELOCATED
Sin 74	Tatopani	6	Bhekasingh	1			389247	3090936	Community should undertake drainage management works RE-ASSESSED BY DS TEAM.
Sin 75	Tatopani	6	Bhumachaur			1	393670	3089889	ALL 28 HOUSEHOLDS TO BE RELOCATED, MOSTLY RELOCATED ALREADY. RE-ASSESSED BY DS TEAM
Sin 76	Tatopani	6	BK Tol			1	393564	3089705	ALL 12 HOUSEHOLDS TO BE RELOCATED, ALL MOVED ALREADY. RE-ASSESSED BY DS TEAM.
Sin 77	Tatopani	6	Patikuna			1	389161	3091174	ALL 19 HOUSEHOLDS TO BE RELOCATED. RE-ASSESSED BY DS TEAM.
Sin 78	Tatopani	6	Thangmoche				389248	3090936	2 HOUSEHOLDS TO BE RELOCATED. RE-ASSESSED BY DS TEAM
Sin 79	Tatopani	6	Vekashing						4 HOUSEHOLDS TO BE RELOCATED

Geo-hazard List

Code	Polygon code (NS)	District	Old Ward	Affected Settlements	Category	Impact on Settlement	Affected Households	Risk Rating	Risk Score
Sin 60		Sindhupalchok	5	Salangkatti	2	Partial			
Sin 61		Sindhupalchok	6	Yarmala-Hikudada	3	Partial	18		
Sin 62		Sindhupalchok	6	Mukchyasa	3	Partial	16		
Sin 63		Sindhupalchok	6	Chyasa	3	Partial	4		
Sin 64		Sindhupalchok	7	Langkhola	2	Partial			
Sin 65		Sindhupalchok	1	Ghattekhola	2	Partial			
Sin 66		Sindhupalchok	2	Nadung	3	Partial	3		
Sin 67		Sindhupalchok	4	9Kilo	2	Partial			
Sin 68		Sindhupalchok	4	Koplang	2	Partial			
Sin 69		Sindhupalchok	5	Fugmache	3	Partial	1		
Sin 70		Sindhupalchok	9	Bokchen/Bukchang	1	Partial			
Sin 71		Sindhupalchok	9	Tartung	1	Partial			
Sin 72		Sindhupalchok	9	Maikharka	1	Partial			
Sin 73		Sindhupalchok	3	Liping	3	Partial	70		
Sin 74		Sindhupalchok	6	Bhekasingh	1	Partial			
Sin 75		Sindhupalchok	6	Bhumachaur	3	Partial	28		
Sin 76		Sindhupalchok	6	BK Tol	3	Partial	12		
Sin 77		Sindhupalchok	6	Patikuna	3	Partial	19		
Sin 78		Sindhupalchok	6	Thangmoche	3	Partial	2		
Sin 79		Sindhupalchok	6	Vekashing	3	Partial	4		

DS2 database

Risk Rating & Prioritization

- Risk Rating for all DMG Assessments & main NRA Geo-Hazard Assessment Completed
- All Category 2 and 3 Communities across 15 Districts
- Risk Rating for assessments undertaken under direction of the NRA are outstanding

$R_N = \text{hazard} \times \text{probability} \times \text{risk value} \times \text{vulnerability}$				
Table 1: Risk Number Scoring Components				
Score	Hazard H	Probability P	Risk Value R	Vulnerability V
1	Small failure or erosion	Unlikely (event expected to occur every 1,000 to 10,000 years)	Agricultural land	Little or no effect
2	Moderate failure – small falling blocks or small shallow failures. All plan dimensions ≤ 100 m, depth ≤ 3 m)	Possible (event expected to occur every 100 to 1,000 years)	Infrastructure (roads, footpaths, water supply)	Minor damage (easily recovered)
3	Substantial failure – large falling blocks or large shallow failure. At least one plan dimension > 100 m, depth ≤ 3 m)	Likely (event expected to occur every 10 to 100 years)	Residential, 1 to 5 households	Significant damage (hard to recover)
4	Substantial deep seated failure. Any failure with depth > 3 m	Very Likely (event expected to occur every 2 to 10 years)	Residential, 6 to 10 households	Major damage (cannot be recovered)
5	Major failure (e.g. Jura landslide scale)	Almost Certain (event expected to occur at least once per year)	Residential, > 10 households	Fatalities

Risk Rating & Prioritization

Reference	Settlement	Hazard, H	Probability, P	Risk Value, R	Vulnerability, V	Risk Score	Survey Group	Survey Group Comments
Sin 60	Salangkatti	3	5	3	4	180	29	Cat II - all scores interpreted from report text
Sin 61	Yarmala-Hikudada	4	5	5	5	500	29	Cat III - vulnerability interpreted
Sin 62	Mukchyasa	4	5	5	5	500	29	Cat III - vulnerability interpreted
Sin 63	Chyasa	4	5	3	5	300	29	Cat III - vulnerability interpreted
Sin 64	Langkhola	3	5	5	4	300	29	Cat II - vulnerability interpreted
Sin 65	Ghattekhol	2	5	1	3	30	29	Cat II - vulnerability interpreted
Sin 66	Nadung	3	5	3	5	225	29	Cat III - vulnerability interpreted
Sin 67	9 Kilo	3	5	5	5	375	29	Cat II - vulnerability interpreted
Sin 68	Koplang	2	5	3	3	90	29	Cat II - vulnerability interpreted
Sin 69	Fugmache	4	5	3	5	300	29	Cat III - vulnerability interpreted
Sin 73	Liping	4	5	5	5	500	8	Cat III
Sin 73	Liping	4	5	5	5	500	8	Cat III
Sin 75	Bhumachaur	3	4	5	5	300	28	Cat III - Assessment team did not provide landslide data, all scores interpreted.
Sin 76	BK Tol	2	4	5	5	200	28	Cat III - probability and vulnerability interpreted
Sin 77	Patikuna	3	5	5	5	375	28	Cat III - vulnerability interpreted
Sin 78	Thangmoche			3		0	28	Cat III - Original assessment was Category 1 (no landslide reported), re-assessed by NRA team as Category 3
Sin 79	Vekashing			3		0	NRA	Cat III
Sin 80	Chitmangu	2	5	4	3	120	28	Cat II - vulnerability interpreted

Risk Rating & Prioritization

Referenc	Settlement	Hazard, I	Probability, P	Risk Value, R	Vulnerability, V	Risk Score	Survey Group	Survey Group Comments
Sin 15	Marming	4	3	1	2	24	DMG	Cat II - problems caused by informal mining, no immediate threat to village
Sin 85	Thangsur	3	3	1	3	27	28	Cat II - all scores interpreted, landslides do not affect village directly
Sin 16	Lidhi	2	5	1	3	30	6	Cat II - reclassified by DS team, considered to be geologically safe, but geographically unsafe.
Sin 65	Ghattekhol	2	5	1	3	30	29	Cat II - vulnerability interpreted
Sin 81	Dhumithang	2	5	1	3	30	28	Cat II - vulnerability interpreted, no landslide, erosion by river, houses and infrastructure not directly affected
Sin 88	Yangri	4	3	1	3	36	6	Cat II - slide in gully, not affecting the village
Sin 58	Fulpingkatti	3	5	1	3	45	29	Cat II - all scores interpreted from report text
Sin 136	Geshtole	3	5	1	3	45	29	Cat II - vulnerability interpreted
Sin 95	Bolgaun	2	3	5	2	60	6	Cat II - CLASSIFIED AS LOW HAZARD, MEDIUM RISK
Sin 140	Halayde	4	5	1	3	60	31	Cat II - vulnerability interpreted
Sin 133	Hotomgran	5	4	1	4	80	31	Cat II - landslide not yet affecting village
Sin 68	Koplang	2	5	3	3	90	29	Cat II - vulnerability interpreted
Sin 130	Bagkhor	2	5	3	4	120	29	Cat II - vulnerability interpreted
Sin 80	Chitmangu	2	5	4	3	120	28	Cat II - vulnerability interpreted
Sin 137	Gyabtole	3	5	3	3	135	29	Cat II - vulnerability interpreted
Sin 60	Salangkatti	3	5	3	4	180	29	Cat II - all scores interpreted from report text
Sin 50	Ambathan	3	5	4	3	180	6	Cat II - CLASSIFIED AS LOW RISK

Risk Rating & Prioritization

Key Points

- Around **120 communities are yet to be scored**, mainly from Sindhupalchowk
- **Level of confidence** varies depending on the degree of interpretation required (DMG reports)
- **Conditions on the ground** have probably changed since assessments were undertaken; some have got better, some have got worse
- **Good starting point for local authorities** and others to identify higher risk areas, plan resources and communicate risk in lead up to the monsoon
- Field verification required

DSCWM & DWIDM Interventions

DWIDM						
	District		Settlement	Cat 2	Cat 3	Risk Score
1	Gorkha	Ghyachok	8 Ghansira/DHANSIRA	1		225
2	Gorkha	Sirdibas	2 Ghatekhola		1	400
3	Gorkha	Kerauja	1 Kerauja Gaun 1		1	
4	Kathmandu	Sankaharapur	2 Karki Tol	1		120
5	Lamjung	BANSAR	1,2 DUBAR	1		240
6	Lamjung	Ghermu	1,2 Upalo Chipla		1	500
7	Nuwakot	Betini	7 Golfu Bhanjyang		1	
8	Ramechhap	GunsiBhadaure	2 DODINI (Siran Tol)	1		
9	Rasuwa	Thuman	8 Mendo	1		120
10	Rasuwa	Haku	5 Thulo Haku		1	500
11	Rasuwa	Thuman	8 Tarsa		1	240
12	Rasuwa	Thuman	8 Desyagaun		1	180
13	Sindhupalchowk	Simpal Kavre	5 Mathlo Kalche		1	
14	Sindhupalchowk	Thankpaldhap	5 Ghatera Tol		1	
15	Sindhupalchowk	Bhotenamlang	1 Kafle Tol		1	

DSCWM						
	District		Settlement	Cat 2	Cat 3	Risk Score
1	Chitwan	Darechok	8 Orek	1		225
2	Chitwan	Kaule	9 Guchhibang	1		
3	Dhading	Jharlang	8 THALI	1		135
4	Dolakha	Chankhu	2 Chortendanda	1		120
5	Dolakha	Sahare	8 Tallo Kalleri	1		80
6	Gorkha	Lapu	3 Khanigaun	1		500
7	Gorkha	Gumda	5 Machhakhola/Machh	1		36
8	Gorkha	Sirdibas	7 DEWAL GAUN	1		80
9	Makwanpur	Dhiyal	4 ChureDanda	1		36
10	Rasuwa	Syafu	9 Syaprubesi	1		300
11	Sindhupalchowk	Baruwa	8 Kami Tol	1		
12	Sindhupalchowk	Baruwa	3 Komrang	1		225
13	Sindhupalchowk	Baruwa	1 Daale	1		
14	Sindhupalchowk	Mankha	7 Bhaise	1		
15	Sindhupalchowk	Sindhukot	2 Dhungechhap	1		
16	Sindhupalchowk	Thankpaldhap	5 Tallo Danda Tol	1		
17	Dhading	Jharlang	7 NECRANG TOLE		1	225
18	Dhading	Darkha	8 Dharna TOLE		1	225
19	Kavrepalanchok	NagreGagarche	9 Madale tol		1	625
20	Okhaldhunga	Prapcha	8,9 Keureni		1	300
21	Okhaldhunga	Ragani	3,4 Gairigau		1	400
22	Sindhupalchowk	Golche	4&5 Nimlung		1	

NRA Geologists Activities

- **Assessments continue** - sign-off of 60 further assessments to the current lists by the NRA Technical Committee is pending this week (5th Phase).
- Mainly **single houses** and **integrated settlements**
- Geologists estimate 15 integrated settlements are in the system – many more unofficial

Consortium's Work Plan & Priorities

Implementation Unit

- Monsoon emergency plan with Gaupalikas
- Development of Durable Solutions website & public dashboard with live updates on the relocation process
- Ongoing facilitation of the relocation program

Geohazard Unit

- Continuous geo-hazard assessments of new sites
- Sharing of landslide risk scoring & prioritization with Gaupalikas; Landslide messaging

Policy Unit

- Policy discussions on Cat-2 mitigation works, Integrated Settlements & rights of landless populations through DS-II Steering Committee

दिगो समाधान

— DURABLE SOLUTIONS —

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